quantum gravity and elementary particles. Oppenheimer, by contrast, seemed not to care much in 1958. He believed his own calculations with Snyder but showed no need to push further, no drive for deeper understanding. Perhaps he was tired from the intense battles of the preceding two decades—weapons design battles, political battles, personal battles. Perhaps he was overawed by the mysteries of the unknown. In any event, he would never again contribute answers. The torch was being passed to a new generation. Oppenheimer's legacy would become Wheeler's foundation; and in the U.S.S.R., Landau's legacy would become Zel'dovich's foundation.

In his 1958 Brussels confrontation with Oppenheimer, Wheeler asserted that the Oppenheimer–Snyder calculations could not be trusted. Why? Because of their severe idealizations (Figure 6.3 above). Most especially, Oppenheimer had pretended from the outset that the imploding star has no pressure whatsoever. Without pressure, it was impossible for the imploding material to form shock waves (the analogue of breaking ocean waves, with their froth and foam). Without pressure and shock waves, there was no way the imploding material could heat up. Without heat and pressure, there was no way for nuclear reactions to be triggered and no way to emit radiation. Without outpouring radiation, and without the outward ejection of material by nuclear reactions, pressure, or shock waves, there was no way for the star to lose mass. With mass loss forbidden from the outset, there was no way the massive star could ever reduce itself below 2 Suns and become a cold, dead, neutron star. No wonder Oppenheimer's imploding star had formed a black hole, Wheeler reasoned; his idealizations prevented it from doing anything else!

In 1939, when Oppenheimer and Snyder did their work, it had been hopeless to compute the details of implosion with realistic pressure (thermal pressure, degeneracy pressure, and pressure produced by the nuclear force) and with nuclear reactions, shock waves, heat, radiation, and mass ejection. However, the nuclear weapons design efforts of the intervening twenty years provided precisely the necessary tools. Pressure, nuclear reactions, shock waves, heat, radiation, and mass ejection are all central features of a hydrogen bomb; without them, the bomb won't explode. To design a bomb, one had to incorporate all these things into one's computer calculations. Wheeler's team, of course, had done so. Thus, it would have been natural